

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks available |     |     |       |       |      |
|----------|-----|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (c) | (i)  |  | both will give a lilac flame / same colour flame (because both contain the potassium ions / $K^+$ / potassium)                                                                                                                                                                                                                                                                                                                                                                            |                 |     | 1   | 1     |       | 1    |
|          |     | (ii) |  | <p>add silver nitrate solution / <math>AgNO_3(aq)</math> (1)</p> <p>potassium chloride gives a (white) <b>precipitate</b> whereas potassium sulfate gives no precipitate / only the chloride gives a (white) <b>precipitate</b> (1)</p> <p>OR</p> <p>add barium chloride solution / <math>BaCl_2(aq)</math> (1)</p> <p>potassium sulfate gives a (white) <b>precipitate</b> whereas potassium chloride gives no precipitate / only the sulfate gives a (white) <b>precipitate</b> (1)</p> |                 | 1   | 1   | 2     |       | 2    |
|          |     |      |  | Question 4 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 6               | 4   | 2   | 12    | 1     | 6    |